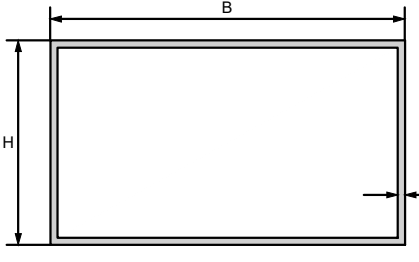


NOMINAL DUCTWORK WEIGHTS - RECTANGULAR

Air Duct: Non-Insulated

Declaration of weight in kg/m

B (mm)	s = 0.75 mm					s = 0.88 mm				
	200	224	250	280	315	355	400	450	500	
200	6.1	6.5	6.9	8.6	9.2	10.0	10.8	11.7	12.6	
224	-	6.9	7.3	9.1	9.7	10.4	11.2	12.1	13.0	
250	-	-	7.7	9.5	10.1	10.9	11.7	12.6	13.5	
280	-	-	-	10.1	10.7	11.4	12.2	13.1	14.0	
315	-	-	-	-	11.3	12.0	12.8	13.7	14.6	
355	-	-	-	-	-	12.8	13.6	14.5	15.4	
400	-	-	-	-	-	-	14.4	15.3	16.2	
450	-	-	-	-	-	-	-	16.2	17.1	
500	-	-	-	-	-	-	-	-	18.0	



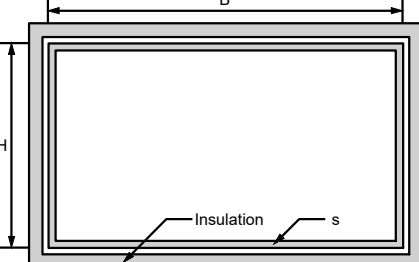
Values for the calculation:
Density air duct: Steel plate $\rho_p = 7,850 \text{ kg/m}^3$
Weights for flanges and duct bracing were estimated.

B (mm)	s = 1.00 mm					s = 1.13 mm					s = 1.5 mm				
	560	630	710	800	900	1,000	1,120	1,250	1,400	1,600	1,800	2,000	2,240	2,500	2,800
200	15.5	16.9	18.6	20.4	22.5	24.5	30.4	33.4	36.9	41.5	46.1	50.7	62.3	68.9	76.5
224	15.6	17.4	19.1	20.9	22.9	25.0	31.0	34.0	37.5	42.1	46.7	51.3	62.9	69.5	77.1
250	16.5	18.0	19.6	21.4	23.5	25.5	31.6	34.6	38.1	42.7	47.3	51.9	63.5	70.2	77.8
280	17.1	18.6	20.2	22.0	24.1	26.1	32.3	35.3	38.7	43.4	48.0	52.6	64.3	70.9	78.6
315	17.9	19.3	20.9	22.8	24.8	26.8	33.1	36.1	39.6	44.2	48.8	53.4	65.2	71.8	79.5
355	18.7	20.1	21.7	23.6	25.6	27.7	34.0	37.0	40.5	45.1	49.7	54.3	66.2	72.8	80.5
400	19.6	21.0	22.7	24.5	26.5	28.6	35.1	38.1	41.5	46.1	50.7	55.4	67.4	74.0	81.6
450	20.6	22.0	23.7	25.5	27.6	29.6	36.2	39.2	42.7	47.3	51.9	56.5	68.6	75.3	82.9
500	21.6	23.1	24.7	26.5	28.6	30.6	37.4	40.4	43.8	48.4	53.0	57.7	69.9	76.5	84.2
560	22.9	24.3	25.9	27.8	29.8	31.8	38.7	41.7	45.2	49.8	54.4	59.0	71.4	78.1	85.7
630	-	25.7	27.3	29.2	31.2	33.3	40.4	43.4	46.8	51.4	56.0	60.7	73.2	79.9	87.5
710	-	-	29.0	30.8	32.9	34.9	42.2	45.2	48.7	53.3	57.9	62.5	75.3	81.9	89.5
800	-	-	-	32.7	34.7	36.7	44.3	47.3	50.7	55.4	60.0	64.6	77.6	84.2	91.8
900	-	-	-	-	36.7	38.8	46.6	49.6	53.0	57.7	62.3	67.3	80.1	86.7	94.4
1,000	-	-	-	-	-	40.8	48.9	51.9	55.4	60.0	64.6	69.2	82.7	89.3	96.9
1,120	-	-	-	-	-	-	51.7	54.7	58.1	62.7	67.3	72.0	85.7	92.4	100.0
1,250	-	-	-	-	-	-	-	57.7	61.1	65.7	70.3	75.0	89.0	95.7	103.3
1,400	-	-	-	-	-	-	-	-	64.6	69.2	73.8	78.4	92.9	99.5	107.2
1,600	-	-	-	-	-	-	-	-	-	73.8	78.4	83.0	98.0	104.6	112.3
1,800	-	-	-	-	-	-	-	-	-	-	83.0	87.6	103.1	109.7	117.4
2,000	-	-	-	-	-	-	-	-	-	-	-	92.3	108.2	114.8	122.5
2,240	-	-	-	-	-	-	-	-	-	-	-	-	114.3	120.9	128.6
2,500	-	-	-	-	-	-	-	-	-	-	-	-	-	127.6	135.2
2,800	-	-	-	-	-	-	-	-	-	-	-	-	-	-	142.9
3,150	-	-	-	-	-	-	-	-	-	-	-	-	-	-	160.7

Air Duct: Insulated

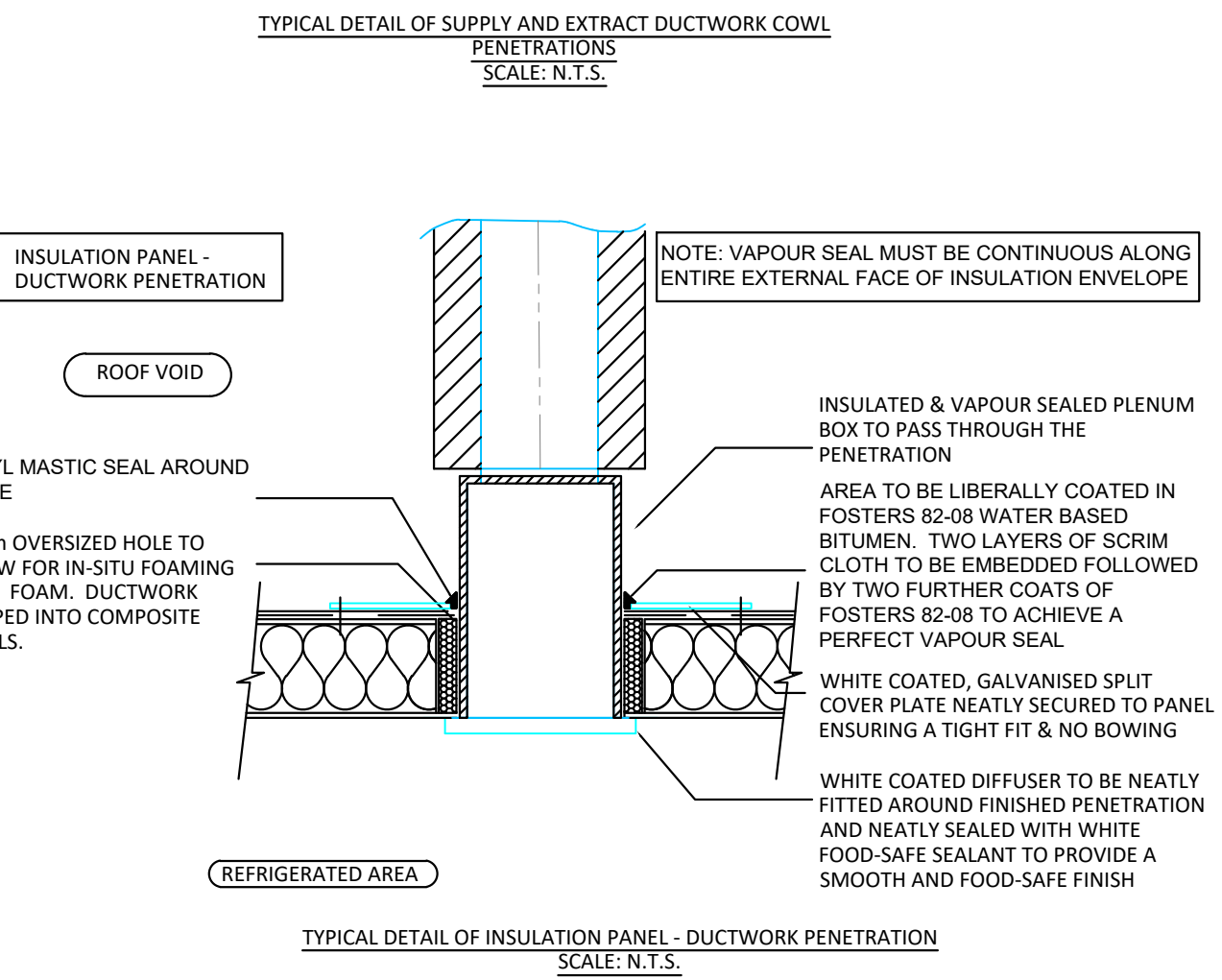
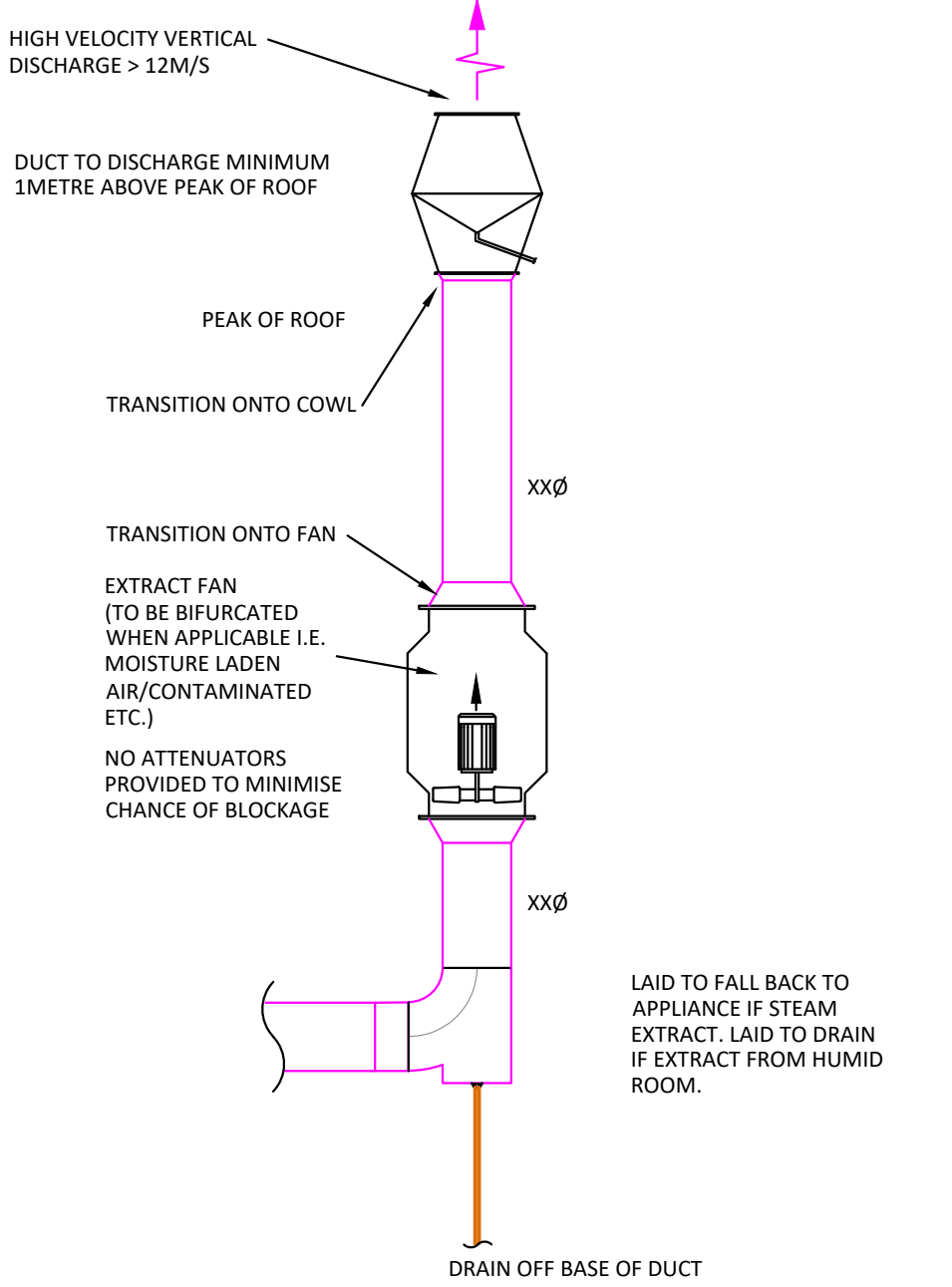
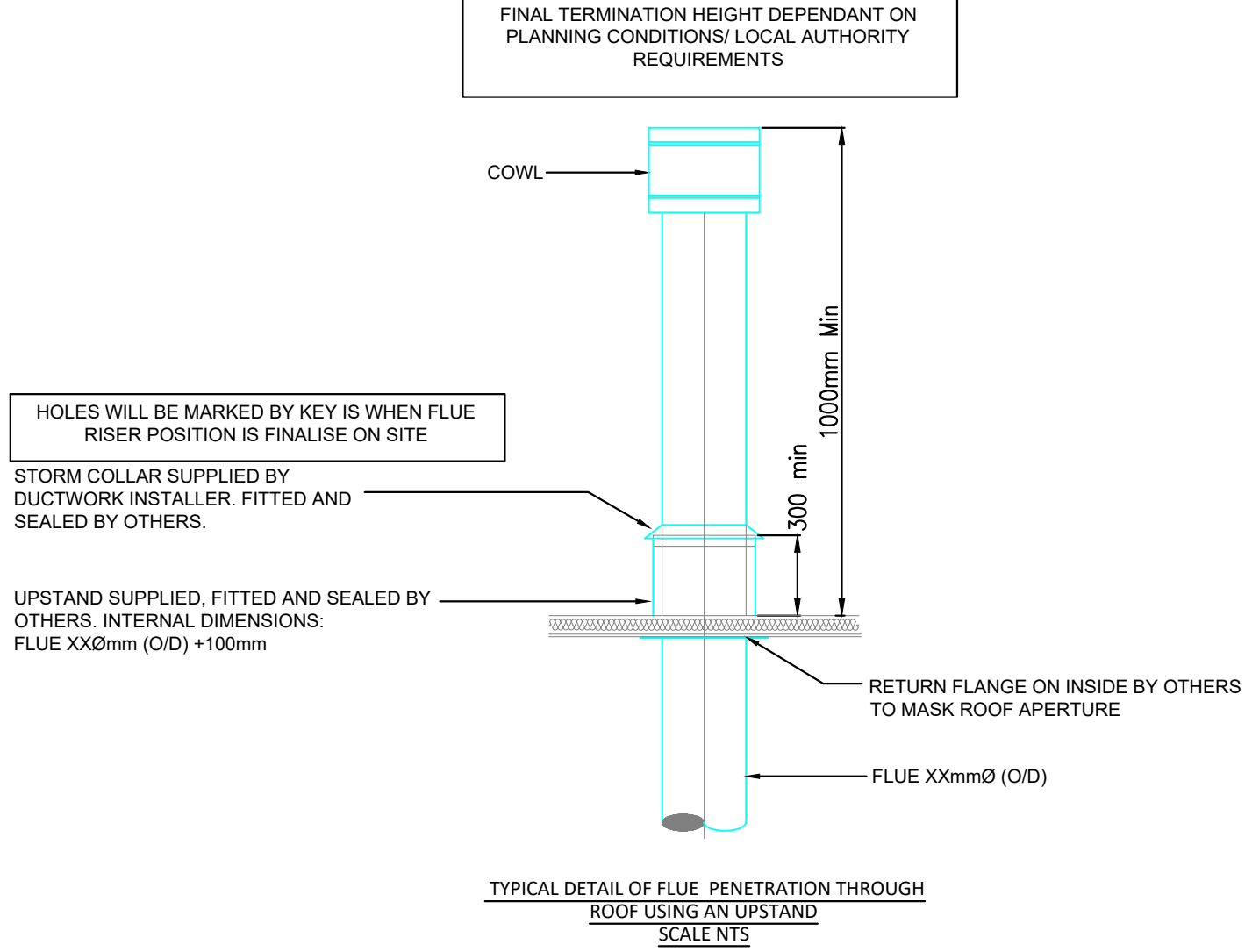
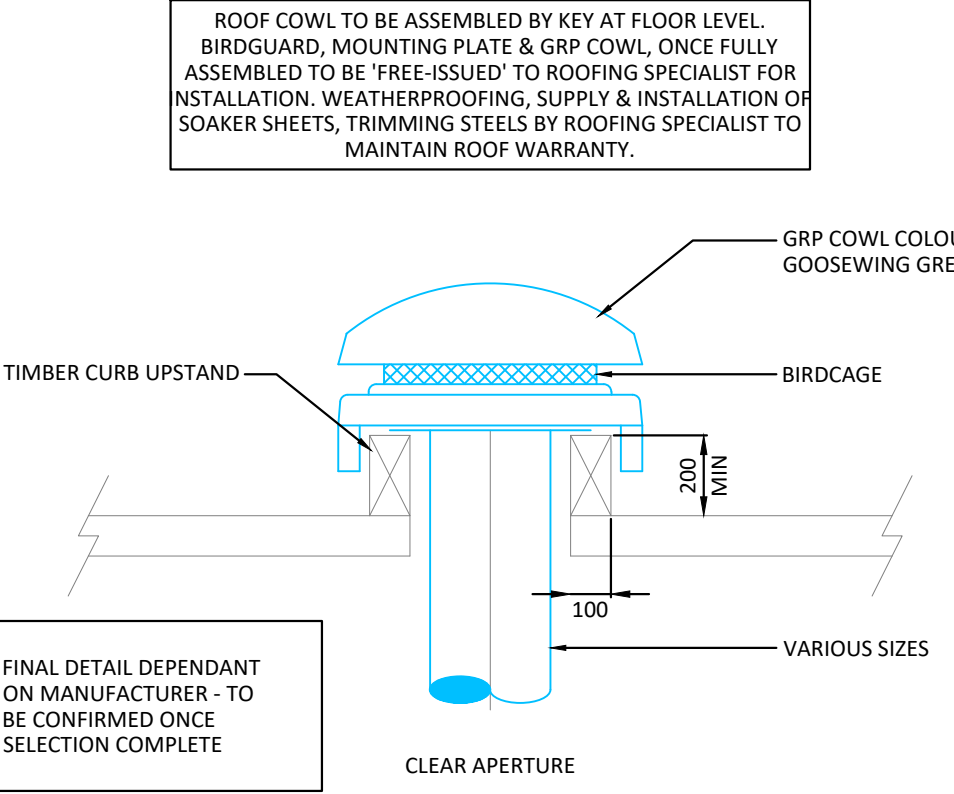
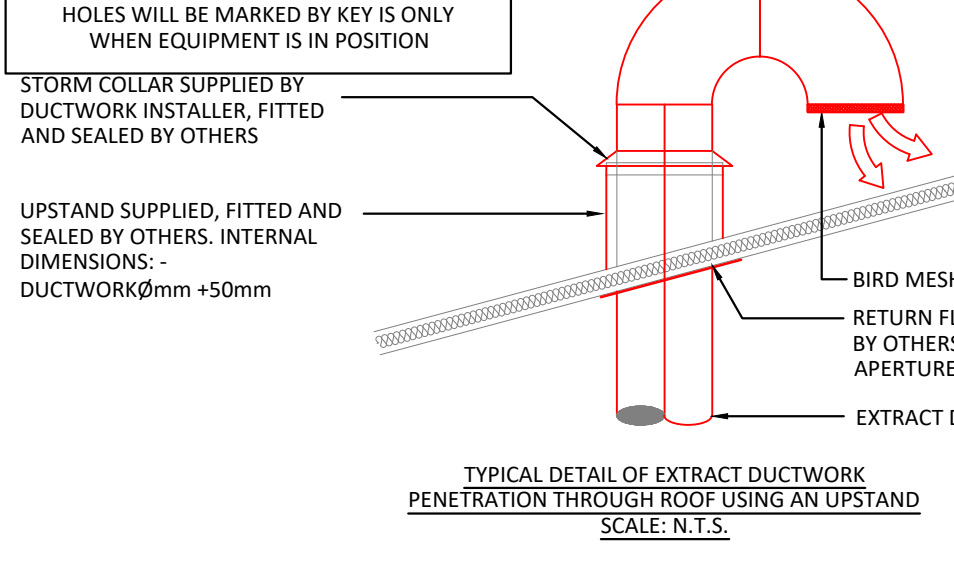
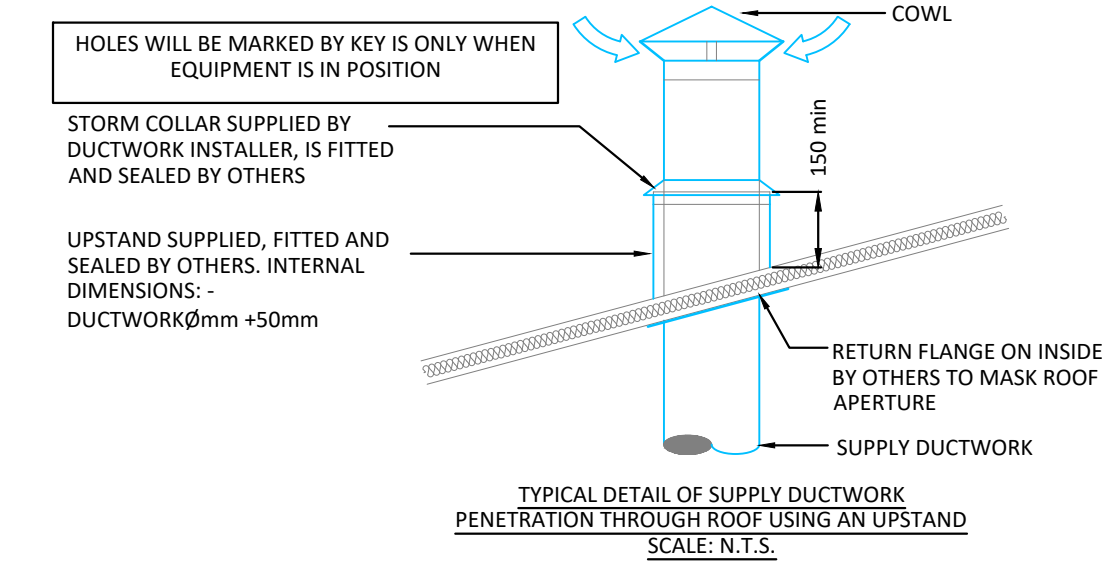
Declaration of weight in kg/m

B (mm)	s = 0.75 mm					s = 0.88 mm				
	200	224	250	280	315	355	400	450	500	
200	9.2	9.9	10.3	12.2	13.1	14.0	15.1	16.3	17.6	
224	-	10.4	10.8	12.8	13.6	14.6	15.7	16.9	18.1	
250	-	-	11.4	13.4	14.3	15.3	16.3	17.6	18.8	
280	-	-	-	14.2	15.0	16.0	17.1	18.3	19.5	
315	-	-	-	-	15.9	16.8	17.9	19.1	20.4	
355	-	-	-	-	-	17.8	18.9	20.1	21.3	
400	-	-	-	-	-	-	20.0	21.2	22.4	
450	-	-	-	-	-	-	-	22.4	23.7	
500	-	-	-	-	-	-	-	-	24.9	

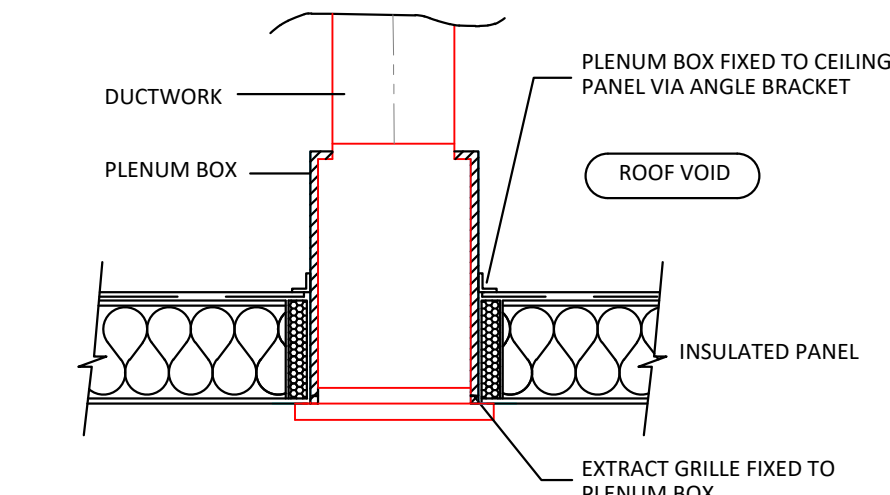


Values for the calculation:
Density air duct: Steel plate $\rho_p = 7,850 \text{ kg/m}^3$
Density insulation: Mineral wool $\rho_i = \text{approx. } 80 \text{ kg/m}^3$
Weights for flanges and duct bracing were estimated.

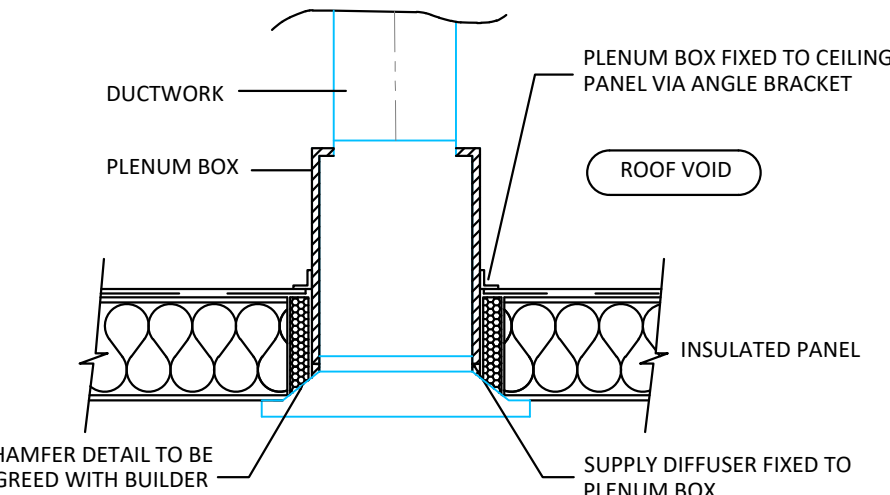
B (mm)	s = 1.0 mm					s = 1.13 mm					s = 1.25 mm				
	560	630	710	800	900	1,000	1,120	1,250	1,400	1,600	1,800	2,000	2,240	2,500	2,800
200	20.9	22.8	24.9	27.3	30.0	32.7	39.4	43.2	47.7	53.5	59.4	65.3	78.4	86.7	96.2
224	21.5	23.4	25.6	28.0	30.6	33.3	40.1	43.9	48.4	54.3	60.1	66.0	79.1	87.4	97.0
250	22.2	24.1	26.2	28.7	31.3	34.0	40.9	44.7	49.1	55.0	60.9	66.8	80.0	88.3	97.8
280	23.0	24.9	27.1	29.5	32.1	34.8	41.8	45.6	50.0	55.9	61.8	67.7	80.9	89.2	98.8
315	24.0	25.8	28.0	30.4	33.1	35.8	42.8	46.6	51.0	56.9	62.8	68.7	82.0	90.3	99.9
355	25.0	26.9	29.1	31.4	34.2	36.8	44.0	47.8	52.2	58.1	64.0	69.9	83.3	91.6	101.2
400	26.2	28.1	30.3	32.7	35.4	38.0	45.3	49.1	53.5	59.4	65.3	71.2	84.8	93.1	102.6
450	27.6	29.5	31.6	34.0	36.7	39.4	46.8	50.6	55.0	60.9	66.8	72.7	86.4	94.7	104.2
500	28.9	30.8	33.0	35.4	38.0	40.7	48.2	52.1	56.5	62.4	68.3	74.2	88.0	96.2	105.8
560	30.5	32.4	34.6	37.0	39.7	42.3	50.0	53.8	58.3	64.2	70.0	75.9	89.9	98.2	107.7
630	-	34.3	36.4	38.9	41.5	44.2	52.1	55.9	60.3	66.2	72.1	78.0	92.1	100.4	110.0
710	-	-	38.6	41.0	43.7	46.4	54.4	58.3	62.7	68.6	74.5	80.4	94.7	103.0	112.5
800	-	-	-	43.4	46.1	48.8	57.1	60.9	65.3	71.2	77.1	83.0	97.5	105.8	115.4
900	-	-	-	-	48.8	51.5	60.0	63.9	68.3	74.2	80.1	86.0	100.7	109.0	118.6
1,000	-	-	-	-	-	54.1	63.0	66.8	71.2	77.1	83.0	88.9	103.9	112.2	121.8
1,120	-	-	-	-	-	-	66.5	70.3	74.8	80.7	86.5	92.4	107.7	116.0	125.6
1,250	-	-	-	-	-	-	-	74.2	78.6	84.5	90.4	96.3	111.9	120.2	129.8
1,400	-	-	-	-	-	-	-	-	83.0	88.9	94.8	100.7	116.7	125.0	134.5
1,600	-	-	-	-	-	-	-	-	-	94.8	100.7	106.6	123.1	131.4	140.9
1,800	-	-	-	-	-	-	-	-	-	-	106.6	112.5	129.4	137.7	147.3
2,000	-	-	-	-	-	-	-	-	-	-	-	118.4	135.8	144.1	153.7
2,240	-	-	-	-	-	-	-	-	-	-	-	-	143.5	151.8	161.4
2,500	-	-	-	-	-	-	-	-	-	-	-	-	-	160.1	169.6
2,800	-	-	-	-	-	-	-	-	-	-	-	-	-	-	179.2
3,150	-	-	-	-	-	-	-	-	-	-	-	-	-	-	201.6



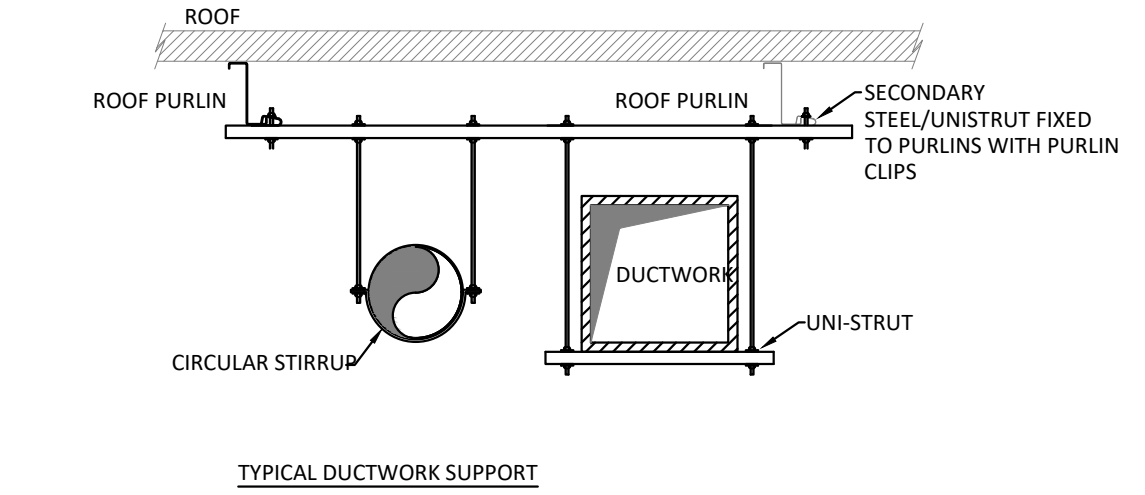
TYPICAL DETAIL OF INSULATION PANEL - DUCTWORK PENETRATION
SCALE: N.T.S.



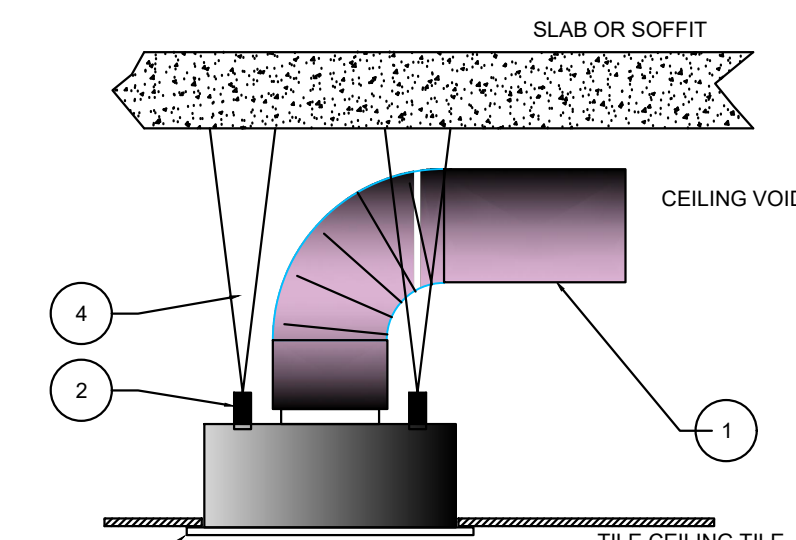
TYPICAL DETAIL OF INSULATION PANEL - STANDARD EXTRACT GRILLE PENETRATION
SCALE: N.T.S.



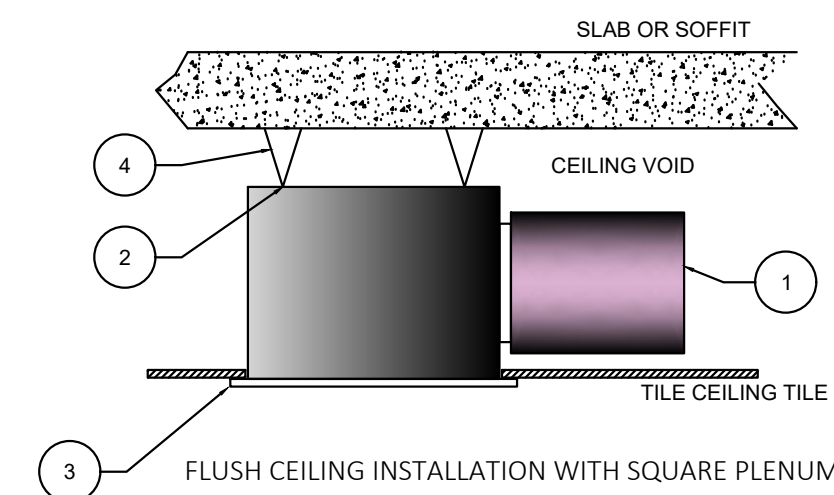
TYPICAL DETAIL OF INSULATION PANEL - STANDARD SUPPLY DIFFUSER PENETRATION
SCALE: N.T.S.



TYPICAL DUCTWORK SUPPORT



- FREELY SUSPENDED INSTALLATION WITH SQUARE PLENUM
- DUCT (FLEXIBLE DUCT FOR FINAL CONNECTION, INSULATED ON SUPPLY DUCT)
 - 4No. SUSPENSION HOLES OR LUGS
 - DIFFUSER FACE (FIXED TO PLENUM)
 - GRIPPLE WIRE & UNI-GRIP SUPPORT BRACKET



- FLUSH CEILING INSTALLATION WITH SQUARE PLENUM
- DUCT (FLEXIBLE DUCT FOR FINAL CONNECTION, INSULATED ON SUPPLY DUCT)
 - 4No. SUSPENSION HOLES OR LUGS
 - DIFFUSER FACE (FIXED TO PLENUM)
 - GRIPPLE WIRE & UNI-GRIP SUPPORT BRACKET

- FLUSH CEILING INSTALLATION WITH SQUARE PLENUM
- DUCT (FLEXIBLE DUCT FOR FINAL CONNECTION, INSULATED ON SUPPLY DUCT)
 - 4No. SUSPENSION HOLES OR LUGS
 - DIFFUSER FACE (FIXED TO PLENUM)
 - GRIPPLE WIRE & UNI-GRIP SUPPORT BRACKET

- FLUSH CEILING INSTALLATION WITH SQUARE PLENUM
- DUCT (FLEXIBLE DUCT FOR FINAL CONNECTION, INSULATED ON SUPPLY DUCT)
 - 4No. SUSPENSION HOLES OR LUGS
 - DIFFUSER FACE (FIXED TO PLENUM)
 - GRIPPLE WIRE & UNI-GRIP SUPPORT BRACKET

NOTES:
1. THIS DRAWING IS THE COPYRIGHT OF KEY IS. IT MUST NOT BE REPRODUCED OR USED FOR ANY PURPOSE WITHOUT PRIOR CONSENT.
2. DO NOT SCALE FROM THIS DRAWING. WORK TO FIGURED DIMENSIONS ONLY. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS STATED OTHERWISE.
3. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH ALL OTHER MECHANICAL, ELECTRICAL & PUBLIC HEALTH DRAWING, SPECIFICATIONS AND SCHEDULES.

VENTILATION SERVICES NOTES

- THIS DRAWING SHALL BE READ IN CONJUNCTION WITH OTHER PROJECT DOCUMENTATION INCLUDING EQUIPMENT SCHEDULES
- FLEXIBLE DUCTWORK SHALL BE USED FOR FINAL CONNECTIONS ONLY. NO FLEXIBLE DUCTWORK SHALL BE MORE THAN 500mm IN LENGTH.
- ALL DUCTWORK SHALL BE MANUFACTURED & INSTALLED IN ACCORDANCE WITH HVAC DW144, DW172 & DW154.
- ALL KITCHEN EXTRACT OR MOISTURE LADEN EXTRACT SYSTEM TO BE INSTALLED IN LINE WITH THE REQUIREMENTS OF DW172.
- DUCTWORK SHALL HAVE ACCESS PANELS IN ACCORDANCE WITH THE REQUIREMENTS OF DW144 AND TR19 AS REQUIRED.
- THERMAL INSULATION THICKNESS SHALL BE IN ACCORDANCE WITH RELEVANT BRITISH STANDARD BS5422
- ALL PLANT AND EQUIPMENT SHALL BE INSTALLED IN ACCORDANCE WITH THE MANUFACTURERS INSTRUCTIONS
- VCD TO BE INSTALLED IN LINE WITH BSRIA BG45/2015 INCLUDING ALL FINAL RUN OUTS
- FRESH AIR & EXHAUST TERMINALS SHALL BE A MINIMUM OF 2m APART. WHERE THE RISER TO ROOF IS LESS THAN THIS THE DUCTWORK SHALL ROUTE HORIZONTALLY AT ROOF LEVEL TO ACHIEVE THE MINIMUM DISTANCE.
- ALL PENETRATIONS TO BE PROVIDED BY THE MAIN CONTRACTOR/ BUILDER
- PLEASE REFER TO FOLLOWING DOCUMENTS FOR DETAILS OF FIRE DAMPERS.
 - KEY-XX-D0C-M-57-001 - FIRE DAMPER TYPICAL DETAIL - ACTIONAIR (SWEGON)
 - KEY-XX-D0C-M-57-002 - FIRE & SMOKE DAMPER TYPICAL DETAIL - ACTIONAIR (SWEGON)

ED2 EDITION 2 - UPDATED FEB 2024

Amendment

YOUR LOGO

Client

Project Title

Drawing Title

Drawn By Checked By Designed By Scale @